

Literature Benchmark Brief: Mulders_2024_RenewEnergy

Title: Nonlinear individual pitch control for load reduction in modern wind turbines

Authors: S. P. Mulders, J. W. van Wingerden (2024)

Venue: Renewable Energy | **DOI:** 10.1016/j.renene.2023.119854

Cluster: Cluster C: Aerodynamics & Control Cliff

Core Theoretical Mechanism & Model Formulation

Aerodynamic lift saturation and asymmetric blade pitch dynamics

Empirical Scale & Experimental Dataset

10-MW wind turbine aeroelastic simulation

Relevance to IEEE TSTE & Sustainable Energy Systems

Documents aerodynamic response non-linearity during turbulent blade pitching

Identified Gaps & Operational Vulnerabilities

Computationally heavy control law requiring real-time sub-second actuation

Role in Manuscript Defense & Reviewer #2 Proof

Reinforces physical reason why stale pitch angles cause cubic over-extrapolation